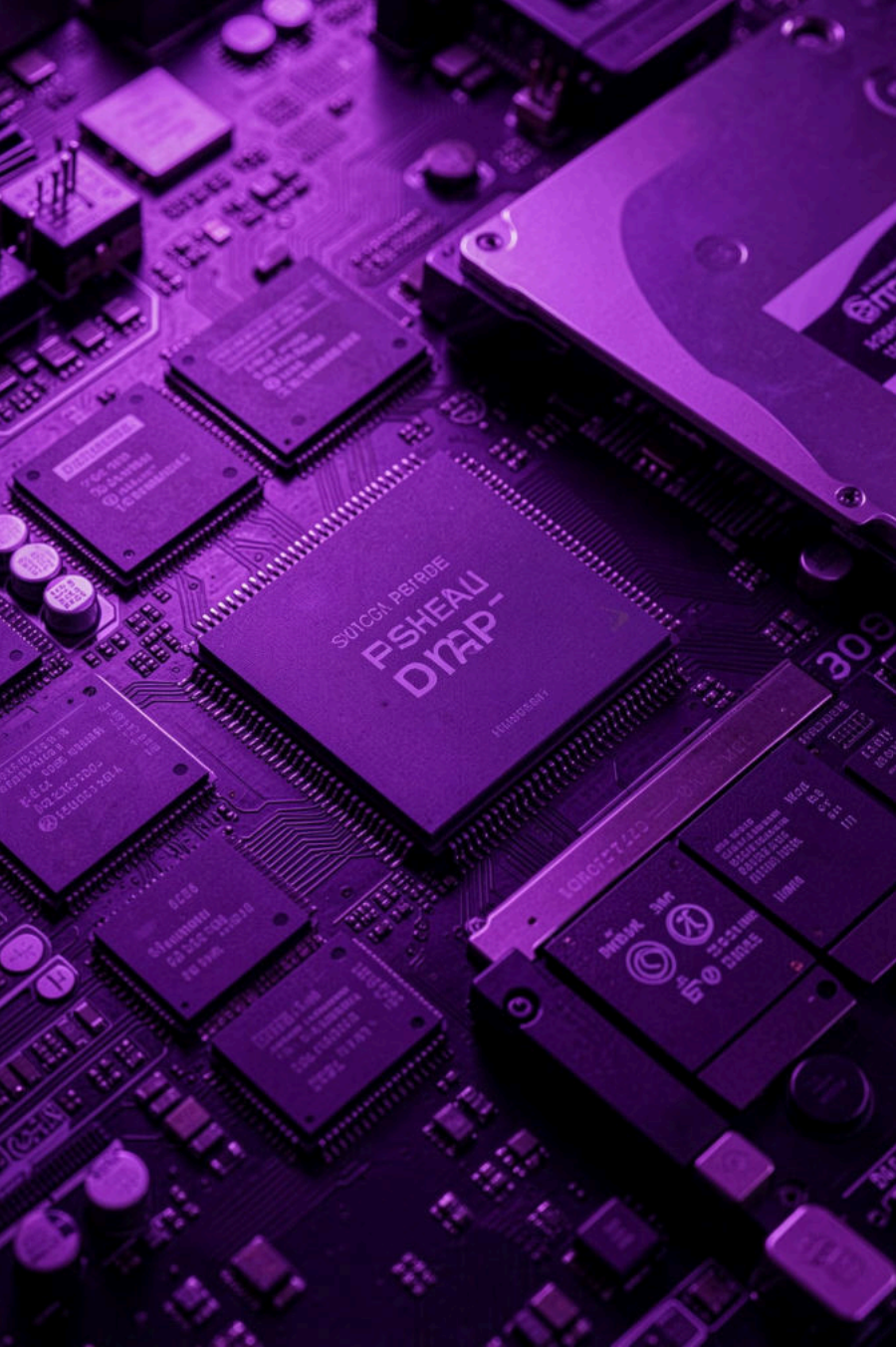




Storage Devices — Computer Memory & Processing

Grade 10 ICT · Module 2.3

MODULE 2.3

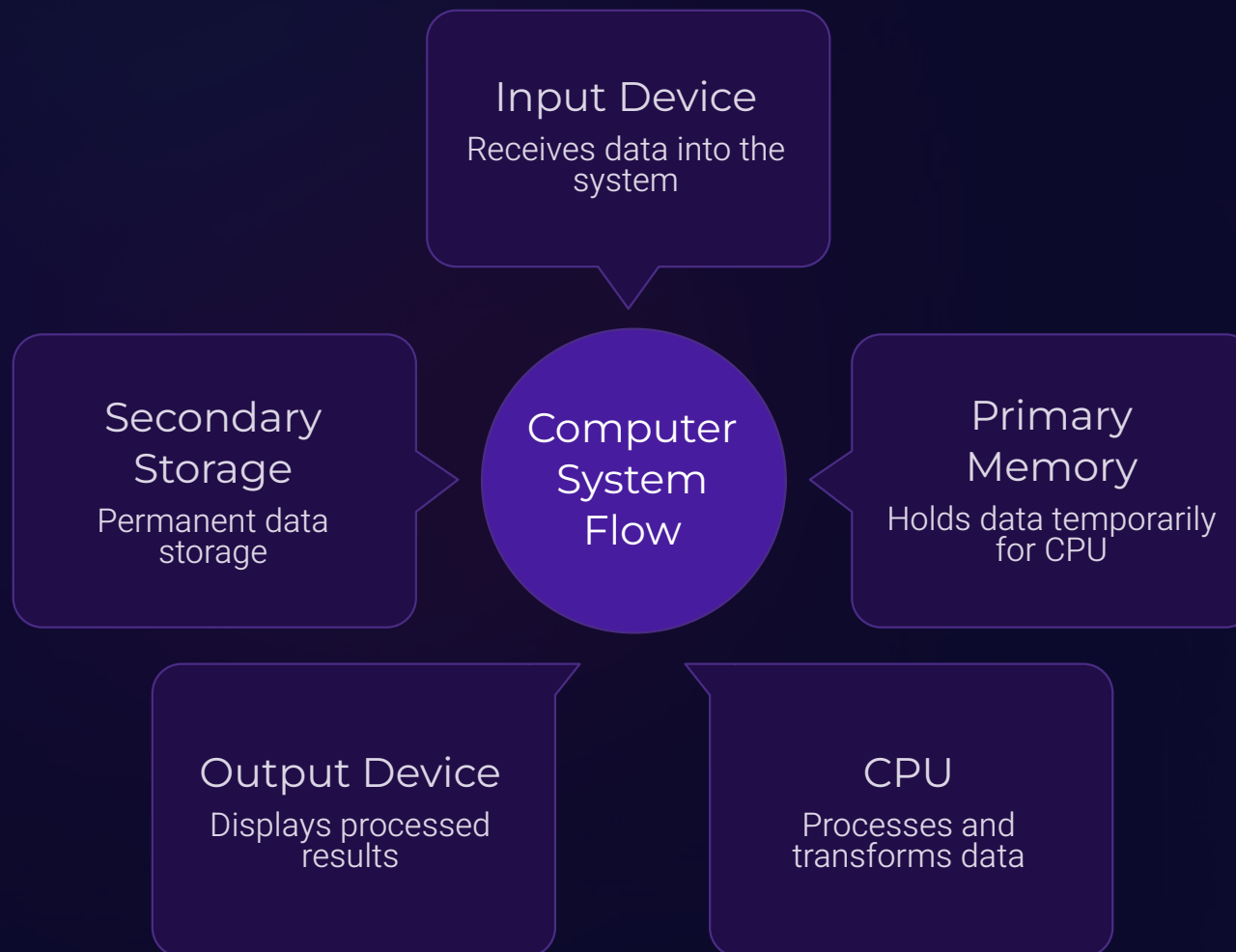
Learning Objectives

- Understand the functions of a computer system
- Identify CPU components (ALU, CU, Registers)
- Understand computer speed measurement
- Differentiate between primary and secondary memory
- Identify various storage devices and their characteristics
- Compare different storage media (Magnetic, Optical, Solid State)

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How a Computer System Works

A computer follows a clear flow: **data enters** via input devices, is **processed** by the CPU, then sent to output devices or saved to secondary storage.



Dotted lines show data flow; solid lines show control signals from the Control Unit to all devices.

The Central Processing Unit (CPU)

The CPU is the **brain of the computer** — it manages the operating system and all application software, executing every instruction given.

Control Unit (CU)

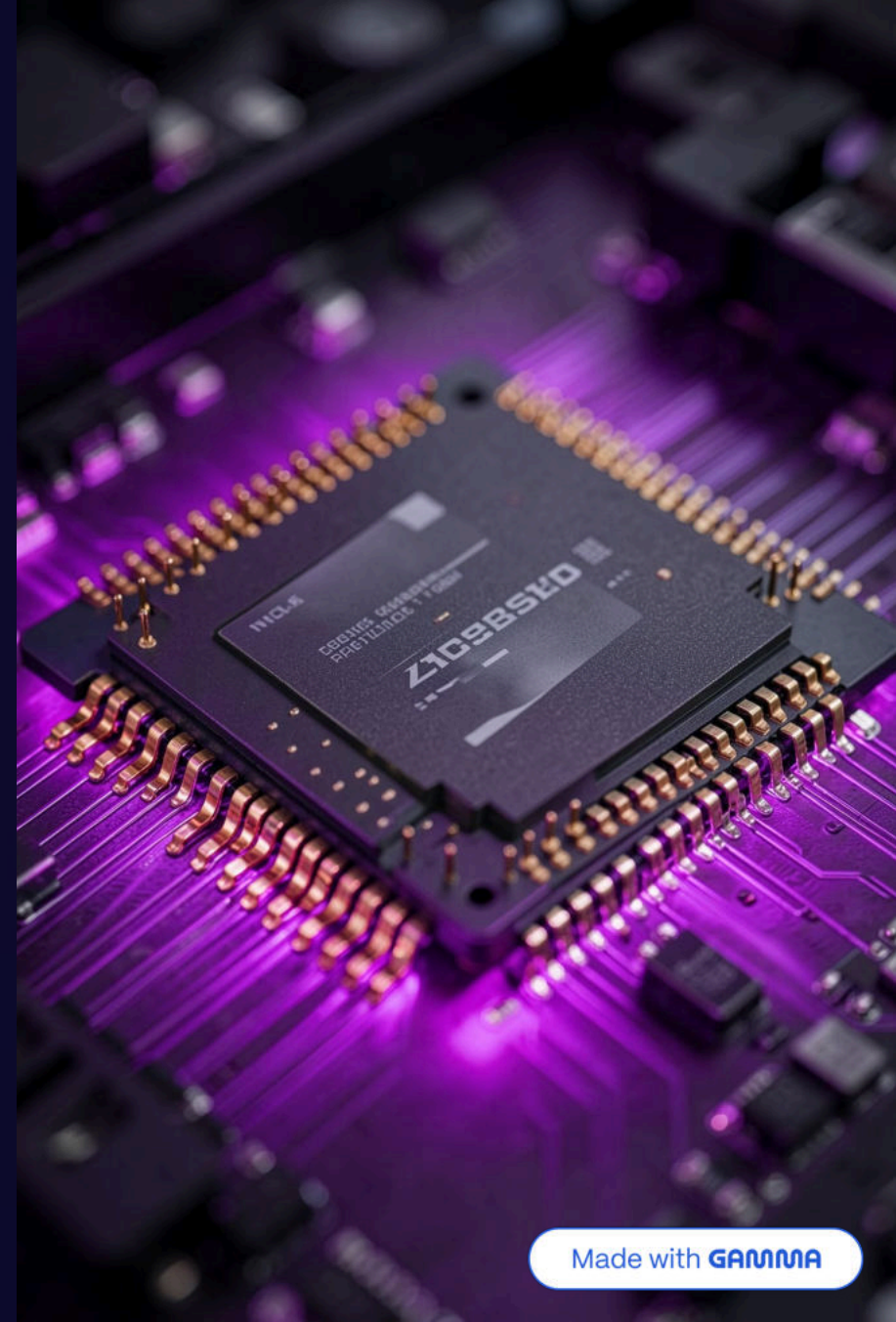
Directs all devices and coordinates operations by sending control signals.

Arithmetic & Logic Unit (ALU)

Performs all mathematical calculations and logical comparisons.

Memory Registers

Ultra-fast temporary storage holding data currently being processed.



Computer Speed



Clock speed is measured in **Hertz (Hz)** – the number of cycles per second. More cycles = faster processing.

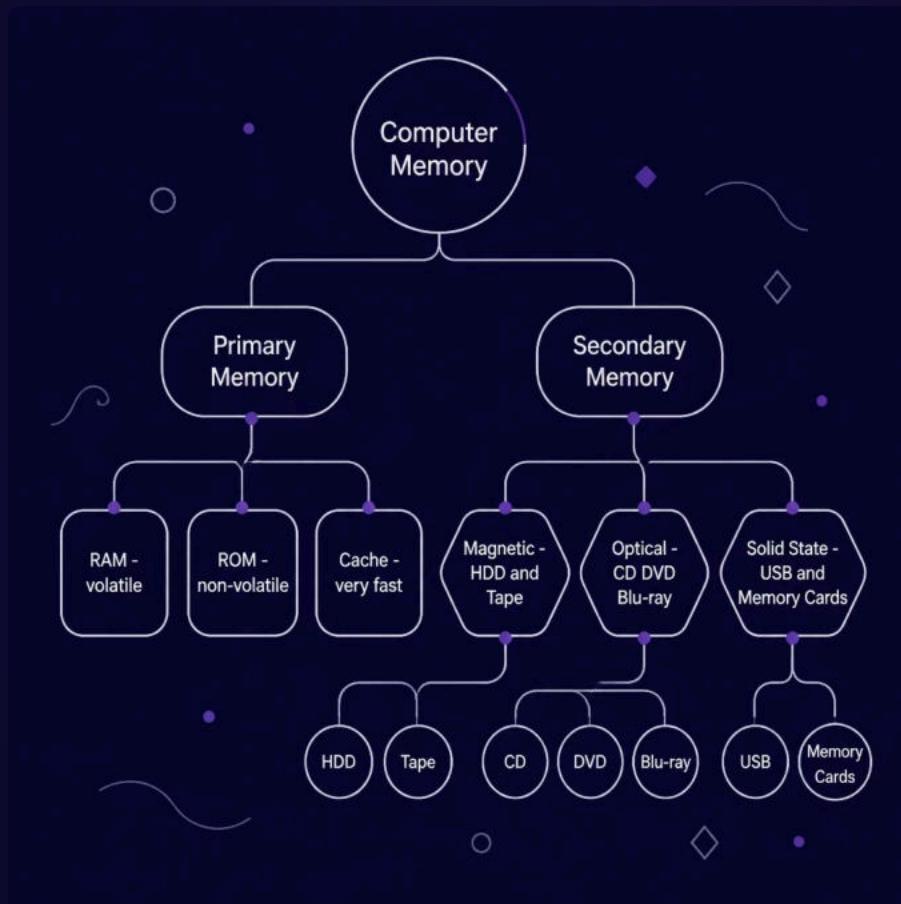
Unit	Abbreviation	Value
Hertz	Hz	1 cycle/sec
Megahertz	MHz	1,000,000 Hz
Gigahertz	GHz	1,000,000,000 Hz



Modern computers run at **2–4+ GHz** – that's billions of operations per second!

Computer Memory — An Overview

Memory devices store data and instructions. They are classified as **primary** (fast, temporary) or **secondary** (permanent, external).



Key Terms

Volatile Memory

Data is **lost** when power is switched off. Example: RAM.

Non-Volatile Memory

Data is **retained** when power is off. Example: ROM, HDD.

Primary Memory — RAM, ROM & Cache



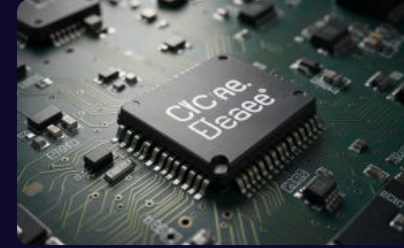
RAM — Random Access Memory

Temporarily stores running programs and data. **Volatile** — erased when powered off. Fast access.



ROM — Read Only Memory

Stores boot instructions (BIOS). **Non-volatile** — data is permanent. Read-only, small capacity.



Cache Memory

The **fastest** memory. Acts as a mediator between CPU and RAM. Stores frequently used data.

📌 **Speed hierarchy:** Cache (fastest) → RAM → Secondary Storage (slowest)

Secondary Storage — Magnetic & Optical Media

Secondary storage holds data **permanently** — it is non-volatile, high-capacity, and cheaper than primary memory.

Magnetic Media

Data stored using **magnetic fields**.

- **HDD:** 250 GB – 4 TB; internal or external; read/write
- **Magnetic Tape:** Used for server backups; large capacity but slow

Optical Media

Data read/written using a **laser beam**.

- **CD:** 650–900 MB (ROM, R, RW)
- **DVD:** 4.7–9.4 GB (ROM, R, RW)
- **Blu-ray:** 25–128 GB for HD video



Solid State Storage Devices

Solid state devices have **no moving parts**, making them fast, durable, and highly portable.

USB Flash Drive

Extremely popular portable storage. Fast read/write speeds. Connects via USB port. Available in various capacities.

Memory Card (SD Card)

Same function as a USB drive but smaller. Used in digital cameras, mobile phones, and gaming devices.

Why Solid State?

No mechanical parts = less prone to damage. Silent operation. Excellent for everyday portable use.

Storage Media — Comparison

Each storage type has strengths suited to different needs.

Feature	Magnetic	Optical	Solid State
Examples	HDD, Tape	CD, DVD, Blu-ray	USB, SD Card
Moving Parts	Yes	Yes (spinning disc)	No
Capacity	Very high (TB)	Medium (GB)	Medium–High
Speed	Fast	Slower	Fastest
Portability	Limited	Good	Excellent
Durability	Moderate	Good	Best

 **Activity:** List advantages and disadvantages of each storage type.



Module 2.3 — Summary & Review



CPU

ALU + CU + Registers.
Speed in MHz/GHz.



Primary Memory

RAM (volatile), ROM
(BIOS), Cache (fastest).



Secondary Storage

Magnetic, Optical, Solid State — permanent storage.

Review Questions

1. Name the three CPU components.
2. Difference between RAM and ROM?
3. What is cache memory used for?
4. Name three secondary storage types.
5. Difference between magnetic and solid state storage?
6. What is volatile memory? Give an example.
7. What is BIOS and where is it stored?



Activity: Identify the ports on a lab computer.

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